Ivan Jutamulia

Senior AI Engineer

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ivanjut

4+ years of experience building production-grade AI systems for real-time sports data delivery at global scale. Rigorous academic background at MIT (B.S. / MEng.) with published research in decision-making modeling. Proven track record deploying low-latency ML pipelines across global football leagues and developing novel deep learning architectures for semantic extraction of tracking data.

SKILLS

- **Programming Languages:** Python (proficient), Rust, JavaScript, Dart, Java, HTML, CSS
- **ML:** PyTorch, TensorFlow, Keras, Scikit-Learn, Transformers and sequence models, Reinforcement Learning, NLP, Computer Vision, Clustering, Regression, Time Series
- **MLOps & Infrastructure:** AWS, Docker, MLflow, Union, GCP, Apache Beam/Dataflow, Pachyderm, CI/CD Pipelines, REST APIs
- **Data:** NumPy, Pandas, Xarray, Polars, SQL, PostgreSQL, MySQL, Firebase/Firestore, Supabase
- **AI Dev:** Claude Code, Cursor, MCPs, Skills
- **Web & App Frameworks:** ReactJS, ExpressJS, VueJS, React Native, Flutter
- **Languages:** English and Mandarin (fluent)

EDUCATION

Massachusetts Institute of Technology — **Cambridge, MA**
Master of Engineering in Computer Science and Artificial Intelligence - GPA: 5.0/5.0 — *June 2021*
B.S. in Computer Science and Engineering, Minor in Statistics and Data Science - Major GPA: 5.0/5.0 — *June 2020*

EXPERIENCE

Genius Sports (Second Spectrum / GeniusIQ) — **Boston, MA (Remote)**
Senior AI Engineer — *September 2021 - Present*

- Derived semantically meaningful insights from spatiotemporal tracking data delivered to soccer clubs, leagues, and governing bodies worldwide, directly informing coaching and scouting decisions, powering broadcast augmentations, and improving officiating fidelity
- Developed a **bi-directional GRU + CRF model** for automatic tracking-derived soccer event detection with 93% F_1
- Built **robust, scalable infrastructure and observability tools** to run **real-time autoeventing system** at < 4 second latency
  - Supported leagues globally: English Premier League, French Ligue 1/2, Belgian Pro League, LigaMX, UEFA, and FIFA
- **Led engineering team** of 15+ and managed production cycle for **Semi-Automated Offsides Technology (SAOT)** adopted across top-tier competitions, increasing offside decision accuracy and reducing review time by > 30 seconds

MIT Sports Lab — **Cambridge, MA**
Undergraduate/Graduate Researcher — *September 2019 - June 2021*

- Developed **Expected Action Value (EAV)** framework to quantify on-court decision-making in the NBA, characterizing and leveraging missed opportunities to evaluate player/team strategy and execution; published in *Journal of Sports Analytics (2025)*
- Utilized **deep learning** approaches to quantify pass and shot difficulty - 0.89 and 0.63 ROC-AUC for respective models
- Integrated into a coaching and analytical tool with full **visualization capability** in collaboration with the San Antonio Spurs

Vim — **San Francisco, CA**
Software Engineering Intern — *January 2020*

- Developed a browser extension using **OCR and DOM manipulation** to extract EHR data and inject real-time provider referral recommendations, reducing manual lookup time for clinicians

Second Spectrum — **Los Angeles, CA**
Machine Learning Intern — *June 2019 - August 2019*

- Launched a tracking data and semantics delivery system for the English Premier League with the AI soccer semantics team
- Trained a **logistic regression classifier** for bisecting passes and integrated into existing live deliverables with 90% F_1
- **Optimized training and evaluation infrastructure** with Pachyderm to speed up model development process by a factor of 20
  - Leveraged framework to retrain and improve the expected goals (xG) model from 0.79 to 0.86 ROC-AUC

MIT Computer Science and Artificial Intelligence Laboratory (LIS group) — **Cambridge, MA**
Undergraduate Researcher — *May 2018 - August 2018*

- Developed a complete **task and motion planning** system for a life-sized PR2 robot to achieve long-horizon tasks
- Integrated a robust computer vision system that could accurately detect objects and their poses with occlusions up to 50%
- Enabled research on reinforcement machine learning and planning in uncertain domains with small real-world datasets

PUBLICATIONS

- Jutamulia, I., & Hosoi, A. E. "Leveraging expected action value to measure on-court decision-making skills in basketball". *Journal of Sports Analytics, 11* (2025)

PROJECTS

Personal AI & Tech News Digest — *In Progress*

- Developing an autonomous agent pipeline that consolidates AI and technology news with **LLM-based reasoning and tool use**
- Architecting a multi-step agentic loop: source discovery → relevance scoring → deduplication → LLM summarization → scheduled delivery
- Exploring **RAG-based personalization** to tailor content weighting based on user reading history and topic preferences